QBS 103 Final Project - Analysis Code

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Submission Notes

Collaborators: none Additional resources used: course materials; ggplot2 documentation (https://ggplot2.tidyverse.org); data source: Mirzakhani et al. (2016), GSE85307, https://pubmed.ncbi.nlm.nih.gov/27841759/ This file makes all figures and the summary table. The figures (folder `figures/`) and the table (`table1.tex`) are uploaded to Overleaf, where the final report is written.

Packages and seed

```
library(ggplot2)    # all plots except the heatmap
library(ggpubr)     # ggarrange(): put plots together in panels A, B, C
library(pheatmap)   # heatmap with clustering and tracking bars
library(knitr)      # kable(): LaTeX table

# Seed so the random jitter in the boxplot is the same every time
set.seed(103)

# Folder for the figure files that go to Overleaf
dir.create("figures", showWarnings = FALSE)
```

Load and wrangle the data

All wrangling starts from the original files provided on Canvas (the three .csv files must be in the same folder as this .Rmd).

```
# Read the csv files
covar <- read.csv("GSE85307_covar.csv", check.names = FALSE)
expr  <- read.csv("GSE85307_geneExpression.csv", check.names = FALSE)
genes_link <- read.csv("GSE85307_geneNames.csv", colClasses = "character")

# The first column of the expression file has the probe IDs: use them as row names
rownames(expr) <- expr[[1]]
expr <- expr[, -1]

# Remove the label before ": " (e.g. "age_years: 31.06" -> "31.06")
for (v in names(covar)[3:ncol(covar)]) {
```

```
  covar[[v]] <- sub("^[^:]*: ", "", covar[[v]])
}

# Change the continuous covariates from text to numbers
num_vars <- c("mat_age_yrs", "delivery_gestAge_wks", "sampleCollection_gestAge_wks",
              "mat_BMI", "mat_vitD")
for (v in num_vars) {
  covar[[v]] <- as.numeric(covar[[v]])
}

# Match covariates to the expression columns using geoID
covar <- covar[match(colnames(expr), covar$geoID), ]

# Check the matching: should print TRUE (every row lines up with its expression column)
all(covar$geoID == colnames(expr))
```

```
## [1] TRUE
```

```
# Initial size of the data: number of probes and samples
dim(expr)
```

```
## [1] 500 157
```

```
# The three genes from Submissions 1 and 2
gene_names <- c("LYN", "JAK2", "PTX3")

# Analysis data frame: one row per woman
dat <- data.frame(
  geoID   = covar$geoID,
  vitD    = covar$mat_vitD,                # continuous covariate of interest
  age     = covar$mat_age_yrs,             # extra continuous covariate 1
  gestAge = covar$delivery_gestAge_wks,    # extra continuous covariate 2
  BMI     = covar$mat_BMI,
  outcome = factor(covar$preg_preeclampsia, levels = c("Normal", "Preeclampsia"))
)

# Derived categorical covariate: obesity status from BMI (obese = BMI >= 30 kg/m^2)
dat$obesity <- factor(ifelse(dat$BMI >= 30, "Obese", "Not obese"),
                      levels = c("Not obese", "Obese"))

# Add the expression of each gene as a column, using the probe-to-gene linker file
for (g in gene_names) {
  probe <- genes_link$PROBEID[genes_link$GENE == g][1]
  dat[[g]] <- as.numeric(expr[probe, ])
}

sum(is.na(dat))                    # should be 0: no samples removed
```

```
## [1] 0
```

```r
table(dat$outcome, dat$obesity)    # group sizes
```

```
##
##                Not obese Obese
##   Normal              79    31
##   Preeclampsia        21    26
```

```r
# One clean theme for all ggplot figures (no main title: figures are described in the
↪  legends)
theme_pub <- theme_classic(base_size = 12) +
  theme(axis.title   = element_text(face = "bold"),
        axis.text    = element_text(colour = "black"),
        legend.title = element_text(face = "bold"))

# Colors used in every figure
col_main    <- "#0072B2"
col_obesity <- c("Not obese" = "#56B4E9", "Obese" = "#D55E00")
col_outcome <- c("Normal" = "#009E73", "Preeclampsia" = "#CC79A7")
```

Table 1. Summary statistics

```r
# Helpers: mean (SD) and n (%) as text
mean_sd <- function(x) {
  sprintf("%.1f (%.1f)", mean(x), sd(x))
}
n_pct <- function(x, level) {
  sprintf("%d (%.1f)", sum(x == level), 100 * mean(x == level))
}

# Split the data by pregnancy outcome
normal <- dat[dat$outcome == "Normal", ]
pe     <- dat[dat$outcome == "Preeclampsia", ]

# Build the table one row at a time (LaTeX symbols are used in the row names)
tab <- data.frame(
  Variable = c("Maternal vitamin D (ng/mL), mean (SD)",
               "Maternal age (years), mean (SD)",
               "Gestational age at delivery (weeks), mean (SD)",
               "Obesity status, n (\\%)",
               "\\quad Not obese (BMI $<$ 30)",
               "\\quad Obese (BMI $\\geq$ 30)"),
  Normal = c(mean_sd(normal$vitD), mean_sd(normal$age), mean_sd(normal$gestAge),
             "", n_pct(normal$obesity, "Not obese"), n_pct(normal$obesity, "Obese")),
  Preeclampsia = c(mean_sd(pe$vitD), mean_sd(pe$age), mean_sd(pe$gestAge),
                   "", n_pct(pe$obesity, "Not obese"), n_pct(pe$obesity, "Obese")),
  Overall = c(mean_sd(dat$vitD), mean_sd(dat$age), mean_sd(dat$gestAge),
              "", n_pct(dat$obesity, "Not obese"), n_pct(dat$obesity, "Obese"))
)
names(tab) <- c("Variable",
                paste0("Normal (n = ", nrow(normal), ")"),
```

Table 1: Characteristics of the study participants by pregnancy outcome. Continuous variables are mean (standard deviation); categorical variables are n (%).

Variable	Normal (n = 110)	Preeclampsia (n = 47)	Overall (n = 157)
Maternal vitamin D (ng/mL), mean (SD)	24.4 (14.5)	19.7 (8.3)	23.0 (13.1)
Maternal age (years), mean (SD)	26.6 (5.1)	25.8 (4.9)	26.4 (5.0)
Gestational age at delivery (weeks), mean (SD)	39.3 (1.0)	37.8 (3.5)	38.8 (2.2)
Obesity status, n (%)			
Not obese (BMI $<$ 30)	79 (71.8)	21 (44.7)	100 (63.7)
Obese (BMI $\geq$ 30)	31 (28.2)	26 (55.3)	57 (36.3)

```
                paste0("Preeclampsia (n = ", nrow(pe), ")"),
                paste0("Overall (n = ", nrow(dat), ")"))

# LaTeX table; escape = FALSE keeps the LaTeX symbols; label gives \label{tab:summary}
table1 <- kable(tab, format = "latex", booktabs = TRUE, escape = FALSE, align = "lccc",
                label = "summary",
                caption = paste("Characteristics of the study participants",
                                "by pregnancy outcome. Continuous variables are",
                                "mean (standard deviation); categorical variables",
                                "are n (\\%)."))

# Save the table for Overleaf (\input{table1.tex}) and show it here
writeLines(table1, "table1.tex")
table1
```

Figure 1. *LYN*: histogram, scatterplot and boxplot

```
# Pearson correlation between vitamin D and LYN
ct <- cor.test(dat$vitD, dat$LYN)
r_txt <- sprintf("%.2f", ct$estimate)
p_txt <- ifelse(ct$p.value < 0.001, "< 0.001", sprintf("= %.3f", ct$p.value))

# Axis label with only the gene name in italics
lyn_lab <- bquote(italic("LYN") ~ "expression (log"[2] ~ "intensity)")

# (A) Histogram with a dashed line at the mean
p_hist <- ggplot(dat, aes(x = LYN)) +
  geom_histogram(bins = 20, fill = col_main, colour = "white") +
  geom_vline(xintercept = mean(dat$LYN), linetype = "dashed", linewidth = 0.8) +
  labs(x = lyn_lab, y = "Number of women") +
  theme_pub

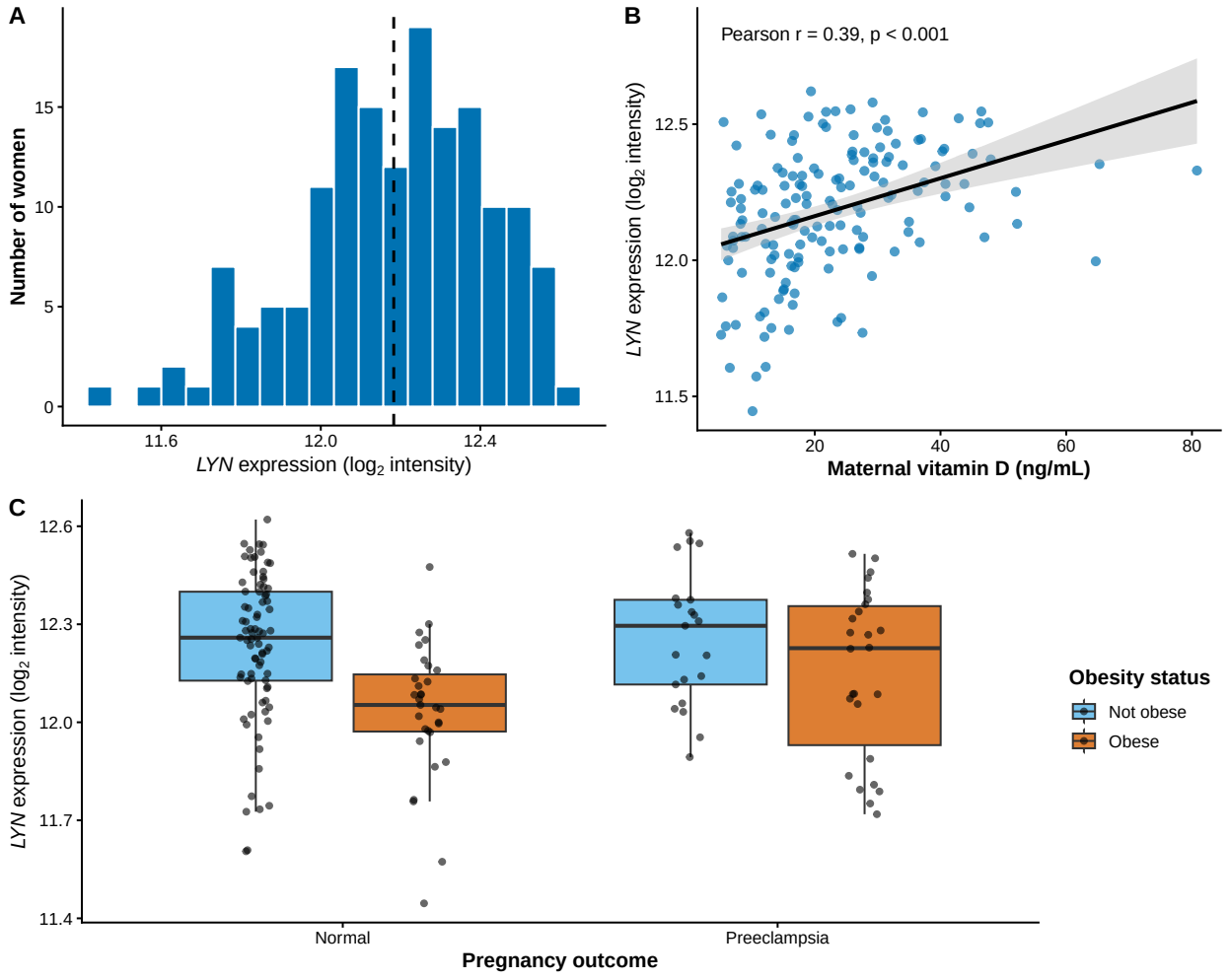
# (B) Scatterplot. The correlation label is placed above the highest point
#     so it cannot overlap any point.
label_y <- max(dat$LYN) + 0.20 * (max(dat$LYN) - min(dat$LYN))

p_scatter <- ggplot(dat, aes(x = vitD, y = LYN)) +
```

```r
  geom_point(colour = col_main, alpha = 0.7, size = 2) +
  geom_smooth(method = "lm", formula = y ~ x, colour = "black", fill = "grey70") +
  annotate("text", x = min(dat$vitD), y = label_y, hjust = 0, vjust = 1, size = 4,
           label = paste0("Pearson r = ", r_txt, ", p ", p_txt)) +
  labs(x = "Maternal vitamin D (ng/mL)", y = lyn_lab) +
  theme_pub

# (C) Boxplot by outcome and obesity with jittered points (random jitter: seed set above)
p_box <- ggplot(dat, aes(x = outcome, y = LYN, fill = obesity)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.8, width = 0.7,
               position = position_dodge(width = 0.8)) +
  geom_point(position = position_jitterdodge(jitter.width = 0.15, dodge.width = 0.8),
             size = 1.4, alpha = 0.6) +
  scale_fill_manual(values = col_obesity, name = "Obesity status") +
  labs(x = "Pregnancy outcome", y = lyn_lab) +
  theme_pub

# Combine into one figure with panels A, B, C
top_row <- ggarrange(p_hist, p_scatter, ncol = 2, labels = c("A", "B"))
fig1 <- ggarrange(top_row, p_box, nrow = 2, labels = c("", "C"))
fig1
```



```r
# Save for Overleaf
ggsave("figures/Figure1_LYN.pdf", fig1, width = 10, height = 8)
```

Figure 2. Heatmap of ten genes

```r
# Genes: the 3 genes above + the 7 most variable named genes in the data
# (picking from the data guarantees that every gene exists and has no missing values)

# Gene name and standard deviation of every probe
probe_gene <- genes_link$GENE[match(rownames(expr), genes_link$PROBEID)]
probe_sd   <- apply(as.matrix(expr), 1, sd)

# Keep probes with a real gene name and some variation, that are not our 3 genes
ok <- !is.na(probe_gene) & probe_gene != "" & probe_gene != "---" &
      !is.na(probe_sd) & probe_sd > 0 &
      !(probe_gene %in% gene_names)

# Sort from most to least variable, keep one probe per gene, take the first 7
cand_probes  <- rownames(expr)[ok][order(probe_sd[ok], decreasing = TRUE)]
cand_genes   <- probe_gene[match(cand_probes, rownames(expr))]
extra_probes <- cand_probes[!duplicated(cand_genes)][1:7]
extra_genes  <- cand_genes[!duplicated(cand_genes)][1:7]

# Probes for our 3 genes + the 7 extra genes
main_probes <- genes_link$PROBEID[match(gene_names, genes_link$GENE)]
heat_genes  <- c(gene_names, extra_genes)
heat_probes <- c(main_probes, extra_probes)
heat_genes
```

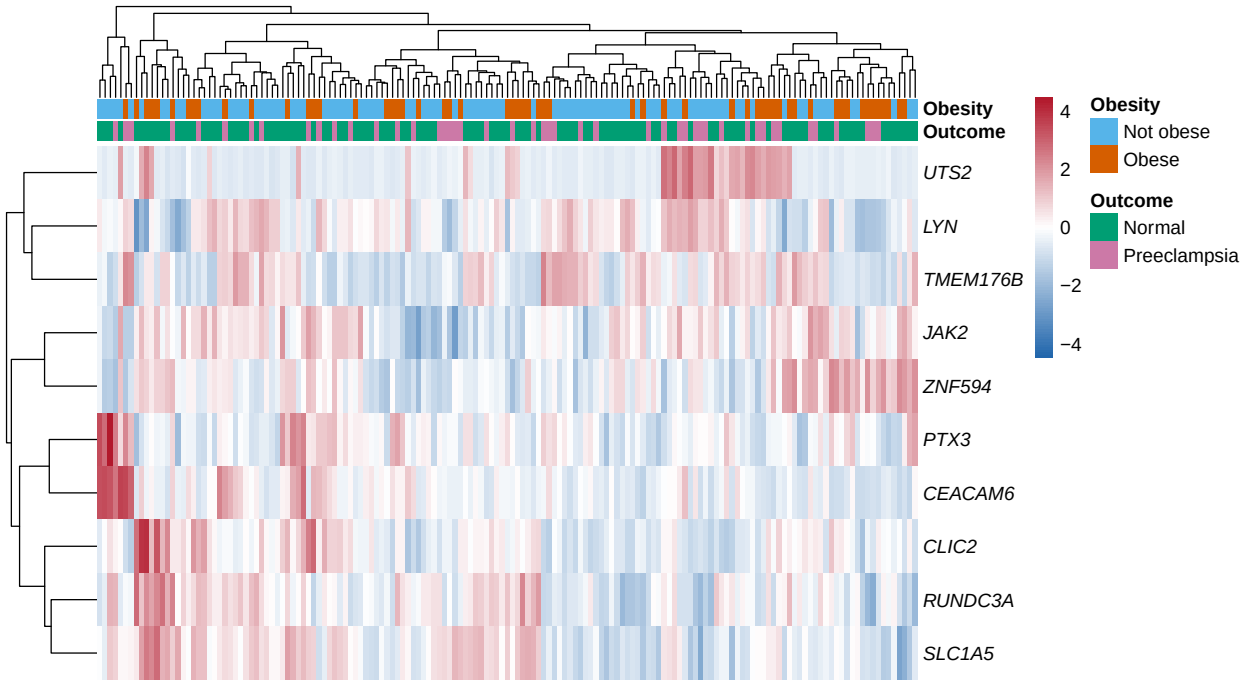
```
##  [1] "LYN"      "JAK2"     "PTX3"     "UTS2"     "RUNDC3A"  "TMEM176B"
##  [7] "CEACAM6"  "CLIC2"    "SLC1A5"   "ZNF594"
```

```r
# Expression matrix: genes in rows, women in columns
heat_mat <- as.matrix(expr[heat_probes, ])
rownames(heat_mat) <- heat_genes

# Tracking bars: one row per woman (row names = sample IDs)
annot <- data.frame(Outcome = dat$outcome, Obesity = dat$obesity)
rownames(annot) <- dat$geoID
annot_cols <- list(Outcome = col_outcome, Obesity = col_obesity)
```

```r
hm <- pheatmap(heat_mat,
               scale = "row",                       # z-score each gene
               # clustering uses the defaults: Euclidean distance, complete linkage
               annotation_col = annot,              # tracking bars
               annotation_colors = annot_cols,
               show_colnames = FALSE,               # 157 sample IDs would be unreadable
               # italic gene names
               labels_row = parse(text = paste0("italic('", heat_genes, "')")),
```

```
              color = colorRampPalette(c("#2166AC", "white", "#B2182B"))(100),
              border_color = NA,
              fontsize = 11)
```




```
# Save for Overleaf
ggsave("figures/Figure2_heatmap.pdf", hm$gtable, width = 10, height = 5.5)
```

Figure 3. Dumbbell plot (additional plot type)

```
# z-score each gene across all women so genes on different scales can be compared
z_mat <- t(scale(t(heat_mat)))

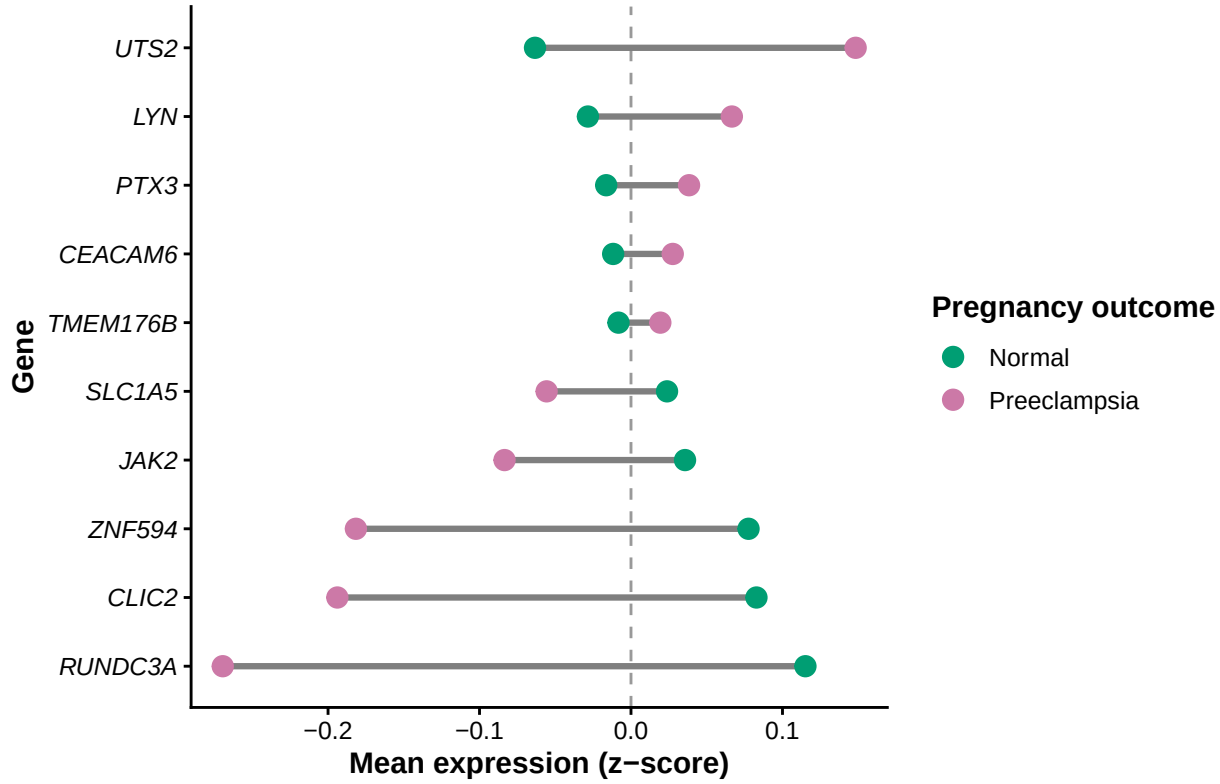
# Mean z-score of each gene in each outcome group
db <- data.frame(
  gene         = heat_genes,
  Normal       = rowMeans(z_mat[, dat$outcome == "Normal"]),
  Preeclampsia = rowMeans(z_mat[, dat$outcome == "Preeclampsia"])
)
db$diff <- db$Preeclampsia - db$Normal

# Order the genes by the difference (largest increase in preeclampsia at the top)
db$gene <- factor(db$gene, levels = db$gene[order(db$diff)])

# Long format for the two colored points
db_long <- data.frame(
  gene    = rep(db$gene, 2),
  outcome = factor(rep(c("Normal", "Preeclampsia"), each = nrow(db)),
```

```
                  levels = c("Normal", "Preeclampsia")),
  mean_z  = c(db$Normal, db$Preeclampsia)
)

fig3 <- ggplot() +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_segment(data = db, aes(x = Normal, xend = Preeclampsia, y = gene, yend = gene),
               colour = "grey50", linewidth = 1.2) +
  geom_point(data = db_long, aes(x = mean_z, y = gene, colour = outcome), size = 3.5) +
  scale_colour_manual(values = col_outcome, name = "Pregnancy outcome") +
  labs(x = "Mean expression (z-score)", y = "Gene") +
  theme_pub +
  theme(axis.text.y = element_text(face = "italic"))   # italic gene names
fig3
```




```
# Save for Overleaf
ggsave("figures/Figure3_dumbbell.pdf", fig3, width = 7, height = 4.5)
```

Numbers used in the report

```
# Data size
nrow(expr)                                  # number of probes
```

```
## [1] 500
```

```r
nrow(dat)                                  # final sample size
```

```
## [1] 157
```

```r
# Table 1 means
c(vitD_normal = mean(normal$vitD), vitD_pe = mean(pe$vitD))
```

```
## vitD_normal     vitD_pe
##    24.37455    19.72468
```

```r
c(gestAge_normal = mean(normal$gestAge), gestAge_pe = mean(pe$gestAge))
```

```
## gestAge_normal     gestAge_pe
##       39.25455       37.81155
```

```r
# Figure 1
mean(dat$LYN)                              # mean LYN expression
```

```
## [1] 12.18291
```

```r
c(r = ct$estimate, p = ct$p.value)      # correlation with vitamin D
```

```
##        r.cor            p
## 3.854955e-01 6.173320e-07
```

```r
tapply(dat$LYN, list(dat$outcome, dat$obesity), median)   # boxplot medians
```

```
##              Not obese    Obese
## Normal        12.25900 12.05300
## Preeclampsia  12.29533 12.22681
```

```r
# Figure 2: the ten genes
heat_genes
```

```
##  [1] "LYN"      "JAK2"     "PTX3"     "UTS2"     "RUNDC3A"  "TMEM176B"
##  [7] "CEACAM6"  "CLIC2"    "SLC1A5"   "ZNF594"
```

```r
# Figure 3: difference in mean z-score (Preeclampsia - Normal), sorted
db[order(db$diff, decreasing = TRUE), c("gene", "diff")]
```

```
##              gene        diff
## UTS2         UTS2  0.21167301
## LYN           LYN  0.09505448
## PTX3         PTX3  0.05471054
## CEACAM6   CEACAM6  0.03935067
## TMEM176B TMEM176B  0.02764771
## SLC1A5     SLC1A5 -0.07949737
## JAK2         JAK2 -0.11917192
## ZNF594     ZNF594 -0.25919431
## CLIC2       CLIC2 -0.27667261
## RUNDC3A   RUNDC3A -0.38465171
```